

AWS EC2 vs. On-Premises Servers: A Comparative Evaluation of Infrastructure Choice for Large Enterprises

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Abstract—As enterprise data continues to grow rapidly and web-based services continue to expand, choosing the right infrastructure has become an important decision for modern organizations. AWS EC2 is often viewed as a flexible solution because it supports fast provisioning, elastic scaling and most importantly it provides dynamic workload management. However, this paper argues that on-premises servers remain highly relevant especially for the large organizations that require stronger control, predictable performance and strict data governance. Thus, this paper compares AWS EC2 and on-premises infrastructure by examining their scaling capability, security control, operational dependency, cost structure and suitability for dynamic web workloads. The paper shows that AWS EC2 do performs well in handling sudden traffic changes but it relies on internet connectivity and third-party cloud providers. These might bring risks for critical enterprise systems. In contrast, on-premises servers may be less flexible in scaling but they offer better ownership, exclusive infrastructure and stronger control over sensitive data. Therefore, this paper suggests that for large companies in regulated sectors such as healthcare, government and finance, on-premises infrastructure can still be a more strategic and reliable choice than full cloud migration.

Index Terms—High-Performance Data Processing (HPDP), AWS EC2, On-Premises Infrastructure, Elastic Scaling, Hybrid Cloud, Data Security

I. INTRODUCTION

As in the era of digitalization, high-performance data processing requires an IT infrastructure that can support rapid data growth, system with higher reliability and changing workload demands. Thus, cloud computing has become widely used because it offers flexibility and faster resource provisioning [1]. However, many large organizations, especially government agencies and healthcare institutions continue to depend on on-premises infrastructure. It is because sensitive data could be protected through direct ownership, strict internal control and stronger governance. This is important in healthcare environments where patient confidentiality and data protection remain major responsibilities for the enterprise [2]. Although virtualized platforms such as AWS EC2 provide elastic scaling and can respond quickly to sudden traffic changes, they also introduce concerns related to internet connectivity, provider

dependency and lesser physical control over the infrastructure [3]. In contrast, on-premises servers may be less flexible in scaling but they offer dedicated infrastructure, predictable performance and stronger control over sensitive systems. Therefore, this paper compares between AWS EC2 and on-premises servers by comparing their technical strengths and operational limitations. The main argument of this paper is that AWS EC2 is suitable for highly dynamic workloads while on-premises infrastructure remains a more strategic and reliable choice for large companies that prioritize data sovereignty, and long-term performance stability. The outline of this paper is organized as follows: Part II describes the architecture and key features of AWS EC2, while Part III discusses the role and importance of on-premises infrastructure. Part IV presents the comparative evaluation between both infrastructure models and Part V concludes the study by justifying why on-premises infrastructure is more suitable for large organizations with critical and sensitive workloads.

II. AWS EC2 ARCHITECTURE AND KEY FEATURES

A. Technology Overview

Amazon Web Services Elastic Compute Cloud (AWS EC2) is one of the primary cloud computing services that offers scalable compute power through virtual servers called instances. These instances can be configured according to different needs like CPU, memory, storage space and networking capacity. EC2 enables organizations to run applications without having to buy and run physical servers, it is a pay-as-you go basis [4]. This can be beneficial for businesses that require quicker provisioning and the ability to allocate resources freely. However, this flexibility comes with the downside that organizations rely on the cloud provider's infrastructure, pricing and service availability.

B. System Architecture

AWS EC2 operates based on worldwide distributed architecture that includes Regions and Availability Zones. They are linked to each other through low latency connections [5]. This structure enables higher availability of applications to multiple locations. A Virtual Private Cloud (VPC) is an

isolated network environment which allows users to manage IP address ranges, subnets, routing and network access control within EC2 [6]. All of these infrastructure remains under the control of the cloud provider and not the organization itself. AWS EC2 also uses Elastic Block Store (EBS) to allow elastic scaling, by separating storage from active computing resources. As shown in Fig. 1, the compute clusters which run virtual machines connect with separate storage clusters via a high-speed data center network (DCN) [7]. This architecture enables EC2 instances to be dynamically added or removed without impacting the stored data. This approach has a drawback where the organization will have less physical control over their services than if they were on premises.

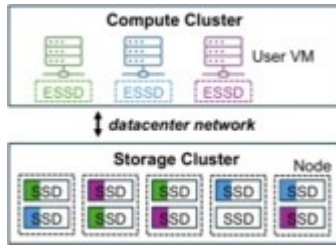


Fig. 1. The storage-compute disaggregated architecture of elastic blockstorage (EBS) [7]

C. Key Performance Features

Auto Scaling enables applications to adjust quickly to changes in demand and minimizes the waste of resources during periods of low traffic volumes [8]. However, the scaling up process is not instant as new virtual machines take time to be launched and be active. Elastic Load Balancing ensures incoming user request could go to another healthy EC2 instance so that no single instance will be overwhelmed by user requests [5]. However, the system still relies on AWS internet connectivity and cloud region availability of the selected region.

D. Case Study : Netflix

Netflix is a good case study to show how AWS EC2 can be used for large, dynamic Web workloads. Using the EC2 service, Netflix can support many instances, minimize latency for viewers in various regions and scale to meet traffic demands during peak hours [9]. This indicates AWS EC2's suitability for global digital businesses with irregular traffic. However, for organizations with sensitive data or high compliance requirements systems which require absolute control of their data, the advantages of elasticity might not outweigh the advantages of direct control, data sovereignty and predictable governance of the infrastructure.

III. ON-PREMISES INFRASTRUCTURE

A. Infrastructure Overview

On-premises infrastructure is the physical servers, storage systems, networking equipment and security equipment that are owned and managed by the organization in its own data

center. The cloud-based platforms would rely on external service providers for computing resources and under their control while on-premises infrastructure would allow the organizations to have full control of the hardware and the software environments by themselves. Although this model requires a larger upfront investment, it is crucial for large enterprises that require long-term control, consistent performance and strict governance over their data. For enterprise systems that manage sensitive or mission-critical data, infrastructure ownership is not only a technical decision but also a strategic decision.

B. Advantages: Security, Control and Sovereignty

The biggest benefit of on-premises servers is it allows organization full control. Organizations can choose how their server machines are set up, where they store data, who has access to them and how to enforce security policies on them. In sectors like healthcare, government and finance where data privacy and compliance are critical, it is particularly important to ensure that such data is well-preserved [10]. Storing data in the organization's environment can help ensure data safety and minimize reliance on external parties [11].

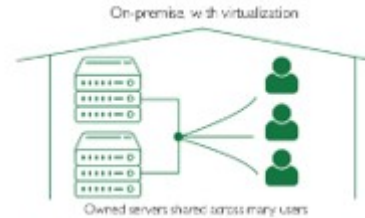


Fig. 2. On-Premises Server Architecture within an Enterprise [12]

The other key benefit is performance predictability. Unlike cloud servers, on-premises servers are only available to the users under the same infrastructure and won't be shared with other cloud tenants as shown in Fig. 2. This means that the 'noisy neighbor' issue is eliminated as any apps that share the same physical infrastructure will have the exact same power and speed. It would not affect the performance of each other. Large enterprises with steady workloads do not need to stretch their computer power frequently. Hence, buying their own servers gives them a reliable, guaranteed speed that never drops. Local availability during internet outage is another benefit of on-premises infrastructure. If there is a problem with internet connectivity the services can still operate within the local network [13]. It is critical for hospitals, government agencies and large companies that cannot afford downtime to their core systems. Thus, while on premise infrastructure may not be as flexible with scaling up quickly, it gives organizations more control, safe and reliability in everyday operation when running workloads.

C. Case Study: KPJ Healthcare & Government Agencies

In Malaysia, healthcare providers such as KPJ Healthcare and various government agencies continue to rely on on-premises infrastructure for important internal systems. For example, Hospital Information Systems manage sensitive patient

records, medical histories, billing details and administrative data that require strict privacy protection and continuous local access. By keeping these systems on-premises, organizations can maintain better control over data storage, access management and compliance monitoring. This shows that on-premises infrastructure is not outdated. It is still a strategic choice for organizations where data ownership and operational continuity are more important than fast cloud scaling.

IV. COMPARATIVE EVALUATION

The decision choosing between AWS EC2 and on-premises infrastructure should not be based only on which platform can scale faster and more flexible. Large enterprises would also consider security control, data ownership, performance stability and dependency risks when deciding on infrastructure. AWS EC2 is highly elastic for dynamic Web workloads, particularly during spikes in traffic. On-premises servers will have better direct control, data sovereignty and more predictable operation for high load sensitive systems. Hence, the comparison is conducted from an enterprise perspective which takes into account not only scalability but also control, compliance and operational dependency.

A. Structured Comparison Table

Table I summarizes the main technical and operational differences between AWS EC2 and on-premises servers.

TABLE I
COMPARISON OF ON-PREMISES INFRASTRUCTURE AND AWS EC2

Comparative Criteria	On-Premises Servers	AWS EC2
Scaling Method	Requires physical upgrades and planned capacity expansion.	Uses Auto Scaling groups to add or remove instances dynamically.
Provisioning Speed	Slower, as installation and configuration may take weeks.	Faster, as virtual instances can be ready for use in minutes.
Performance Predictability	High, as resources are dedicated to one organization and are not shared with other tenants.	Can be high, but performance depends on provider infrastructure conditions.
Security and Control	Provides stronger physical control over servers, network design, access rules, and internal security policies.	Provides strong cloud security features, but still depends on provider infrastructure and isolation.
Data Sovereignty and Compliance	Provides stronger support for strict data privacy and residency because data remains within the organization's environment.	Compliance tools are available, but sensitive data is still hosted in a third-party cloud environment.
Comparative Criteria	On-Premises Servers	AWS EC2
Operational Dependency	Has lower external dependency because systems can operate within the local network even during internet disruption.	Has higher dependency on internet connectivity, AWS service availability, and selected cloud region stability.
Best Suitability	Suitable for large organizations with critical systems, sensitive data, compliance needs, and stable workloads.	Suitable for highly dynamic, public-facing and globally distributed workloads with unpredictable demand.

B. Critical Analysis of Performance Trade-offs

The main strength of AWS EC2 is the ability to provide rapid scaling when there is sudden traffic load. With the use of Auto Scaling and Elastic Load Balancing, the instances may be increased and traffic distributed efficiently, making the cloud service appropriate for web workloads with high unpredictability. At the same time, AWS EC2 is not entirely free from performance risks since the system still depends on internet connectivity, AWS provider, AWS availability in the region and time needed for new instances to start functioning [5]. Large companies that run critical systems will find these as serious limitation. Although on-premises servers have less flexible for elastic scaling, they demonstrate more performance predictability and provide higher levels of control. There is no need to use shared infrastructure and deal with issues related to other users' instances (i.e., the "noisy neighbour" effect). Thus, on-premises server infrastructure can be better in terms of stability, throughput, security and continuous internal access. What is even more important, all data will remain with the organization, ensuring enhanced opportunities for implementing policies related to security and compliance. Hence, while AWS EC2 is more elastic, the latter solution is better for large-scale organizations that prioritize security, compliance, predictable performance and long-term operational stability.

V. CONCLUSION

Overall, this paper revealed that AWS EC2 performs efficiently for elastic web applications. The solution provides easy and fast scaling of resources and helps organizations respond to changes in traffic demand rapidly. However, AWS EC2 still has performance weaknesses related to dependencies on internet connectivity, AWS availability in the selected regions and infrastructure that is managed by third parties. These dependencies may create risks for large organizations that operate sensitive or mission-critical systems. Based on the comparison, on-premises infrastructure remains the stronger choice for large enterprises that prioritize data sovereignty, compliance, predictable performance and direct control. Although on-premises servers require higher initial investment and slower physical scaling, they provide dedicated resources, local ownership and stronger governance over security and data management. Therefore, for sectors such as healthcare, government and finance, on-premises infrastructure is more suitable than full cloud migration because control, reliability and long-term stability are more important than elastic scaling alone.

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